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## Letter Section

# Palaeomagnetic reconnaissance result from the Maubisse Formation, East Timor, and its tectonic implication

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ABSTRACT

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A preliminary palaeomagnetic investigation of the supposedly allochthonous Permian Maubisse Formation in East Timor yielded a mean direction of magnetization of decl. = 140E, incl. = +45 and $\alpha_{95}$ = 13°. This direction is indistinguishable from that of the autochthonous Permian Cribas Formation reported earlier. The apparent agreement between the palaeomagnetism of the two Formations is interpreted to imply that both are autochthonous and have been part of the Australian continental margin since the Permian, and does not support the supposed Asian origin for the Maubisse Formation.

## INTRODUCTION

The Banda Arc system comprises the sinous double chain of islands southeast of the Banda Sea and marginal to northern Australia. The inner arc, flanking the Banda Sea consists of volcanic islands, whereas the outer arc, adjacent to Australia is non-volcanic with formations dating back at least as far as the Permian (Van Bemmelen, 1949).

It is commonly believed (Audley-Charles et al., 1972; Fitch and Hamilton, 1974; Hamilton, 1976; Audley-Charles and Milsom, 1974; Carter et al., 1976; and Chamalaun et al., 1976) that the Banda Arc resulted from the Cenozoic northward drift of Australia and its subsequent collision with an Asian island arc—subduction zone. As the collision may have been as recent as the Pliocene, the region provides an unique opportunity to study the initial deformation history associated with this type of plate convergence.

Timor is the largest island in the Outer Arc chain and renowned for its large-scale overthrust structures. Their presence was initially suggested by Wanner (1913), and since then, overthrusts have become an integral part of

the geological interpretation of Timor (Van Bemmelen, 1949; Grunau, 1953; Gageonnet and Lemoine, 1958; Audley-Charles, 1968; and others). In general, overthrust structures could be regarded as consistent with the hypothesis of plate convergence for the region. However, there is as yet no general agreement regarding the nature, origin and mode of emplacement of the thrust sheets, nor have they all been unanimously identified as such (Lemoine, 1959; Veevers, 1969; Audley-Charles and Carter, 1972; Carter et al., 1976; Grady, 1976).

Following extensive fieldwork in Timor, Audley-Charles and co-workers have developed a series of models for the overthrust history of Timor; the most recent one is that described by Carter et al. (1976). On the basis of their interpretation of field relationships and structural—stratigraphical analysis they divide the geological formations into twenty (para) autochthonous and four allochthonous. One of the most important allochthonous formations is the Permian Maubisse Formation (Audley-Charles, 1968). They suggest that while the Permian and Mesozoic autochthonous formations show strong Australian faunal affinities and may have been part of the Australian continental margin, the allochthonous Permian (i.e. the Maubisse Formation), have more typical tropical faunas and resemble strata in the Sunda Arc, Kalimantan and the Philippines. They speculate therefore that the allochthonous Timor originated as part of an Asian outer arc ridge, perhaps some thousands of kilometres to the north of the Australian margin. Spreading of the marginal Banda Sea, caused the ridge to move away from S.E. Asia, and to be eventually thrust onto the Australian continental margin in the Middle Pliocene.

A different view is expressed by Fitch and Hamilton (1974) and Hamilton (1976). While accepting the basic collision hypothesis, Timor is viewed as a chaotic imbricated tectonic melange, created by a southward moving island-arc system ramping up the Australian continental shelf. No large horizontal displacements are postulated and most formations are presumed to be Australian in origin.

Recent field mapping by Grady (1976) in the type section of the Maubisse Formation, has failed to confirm the existence of horizontal thrust planes. Rather the contact between the Maubisse and other formations was mapped by him as a steep angle fault.

To delineate the overthrust history of Timor in detail is clearly of critical importance to an understanding of the plate convergence process as typified by the Banda Arc systems. In particular it is necessary to identify unambiquously those formations that are allochthonous and their spatial relationship to Australia since the Permian.

The palaeomagnetic method offers an alternative method for testing the various hypotheses. It is based on the proposition that if the Maubisse Formation is allochthonous, either in the sense of a large-scale overthrust, as proposed by Carter et al. (1976) or as part of a tectonic mélange (Fitch and Hamilton, 1974), it is unlikely that its palaeomagnetic direction would be

consistent with that of formations of a similar age either in Timor or Australia.

Consistent directions would argue against an allochthonous origin in either sense and imply an authochthonous origin. The palaeomagnetic results reported here favour the autochthonous interpretation. While the sampling density is low and obviously needs extending, the results are nevertheless presented at this stage because of their bearing on the overthrust hypothesis and because the political instability of the region may prevent a more extensive sampling program for some time.

## MAUBISSE FORMATION AND PALAEOMAGNETISM

The Maubisse Formation is one of four Formations originally mapped by Audley-Charles (1968) as wholly allochthonous. The type section is the large cliffs southwest of the village of Maubisse in the Central highlands of East Timor (see Figs. 1 and 2). Here the Formation consists of well-bedded dense limestone and massive reefs. Chocolate red, dark and grey green shales occur as thin developments within the limestone. Near Mt. Ramelau there occurs a sequence of strongly coloured metamorphosed basic volcanics. Outside the type section the Maubisse Formation is fairly widespread forming distinctive features in the landscape.

Audley-Charles (1968) regards the Maubisse as allochthonous and interprets the field relations to indicate that it rests tectonically on the autochthonous Aitutu (Triassic) and Wai Luli (Jurassic) Formation. The limestones produced a rich fauna, which suggests an age ranging from Early to Late Permian (Audley-Charles, 1968).

Audley-Charles (1968) interpreted the depositional environment to be characterized by shallow, clear warm water with occasional volcanic activity and notes the apparent paradox between the Permian warm water reef limestone in Timor and the Permian glacial deposits in Northern Australia (David, 1950; Veevers and Wells, 1961; Crowell and Frakes, 1971). This led to the suggestion that the Maubisse Formation was formed at the northern margin of the Westralian Geosyncline perhaps as far away as Sulawesi or Kalimantan (Audley-Charles, 1968; Carter et al., 1976).

In this preliminary collection, sampling was confined to the red shales in the type section. The red beds weather readily and good outcrops are rare. Four sites were sampling along the road to Same. (See Fig. 2 for sampling sites). The lithologies sampled include red micaceous siltstones, red crinoidal limestone and chocolate red shales. Outcrops are not large and at each site three to four samples were obtained covering a vertical thickness of 3—4 metres. Since each site consists of different lithologies, they probably represent different time zones. In Fig. 4 samples from the same site are connected, but as the sampling density is low, no attempt has been made to compute individual site means. The results are summarized in Fig. 4. Figure 3A shows

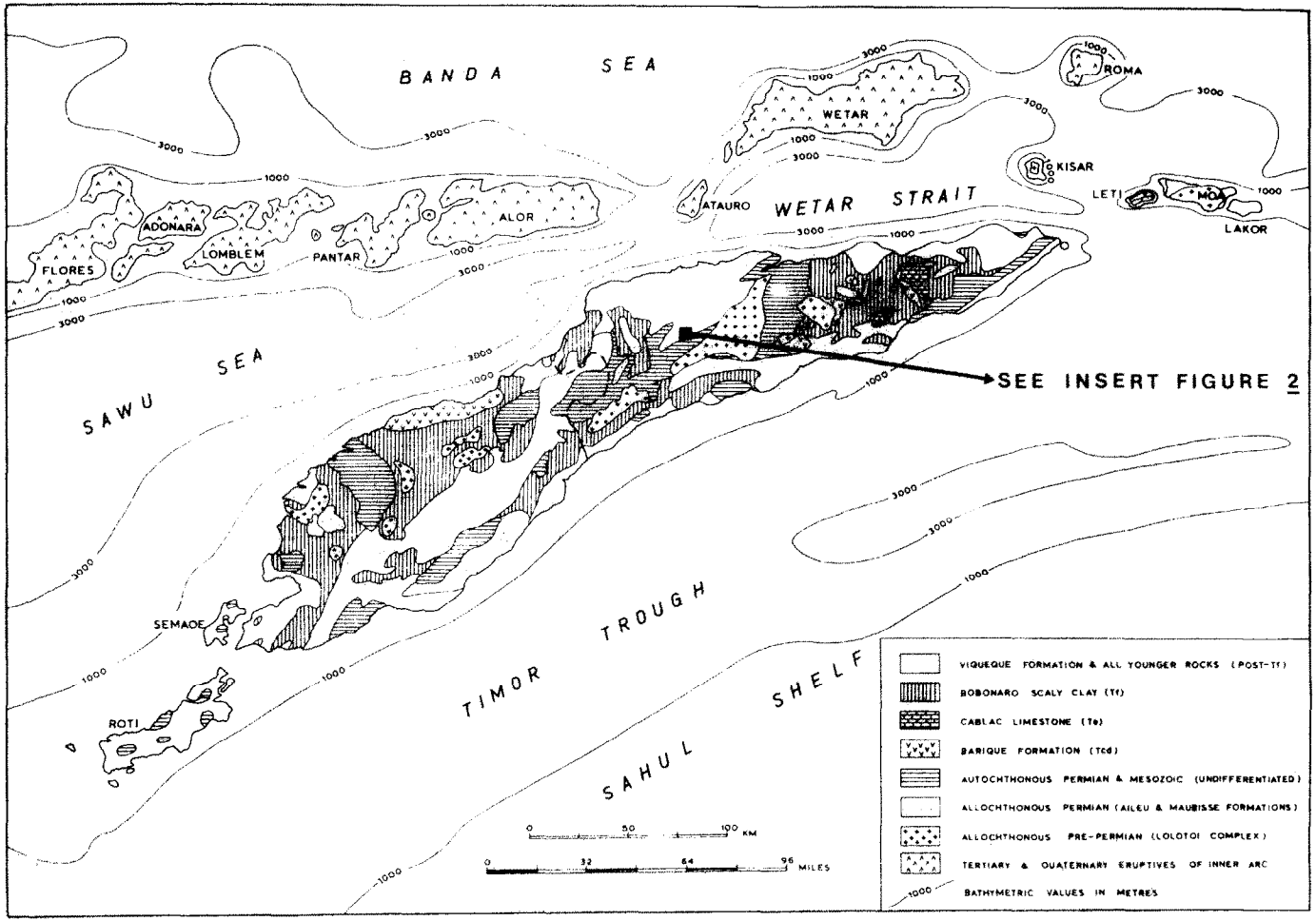



Fig. 1. Shows the general outline of the geology of Timor, according to Audley-Charles (1968), and locality of Fig. 2.

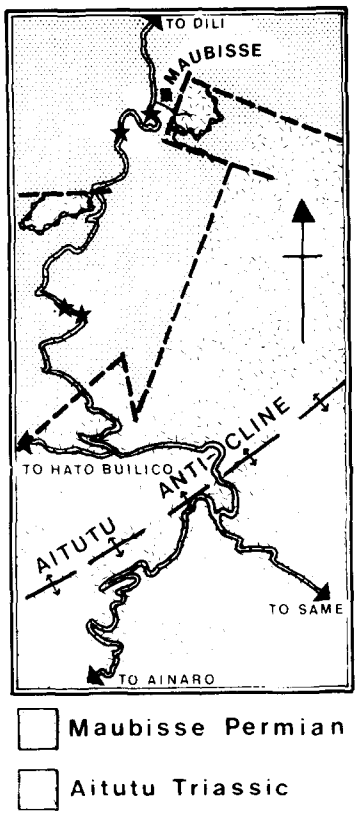



Fig. 2. Generalized geology of the sampling area according to Grady (1976). Sampling sites are shown by stars.

the N.R.M. directions with respect to the present horizontal. The directions show considerable scatter with predominantly negative inclinations. Correction to the palaeohorizontal (Fig. 3B) does not improve the scatter. Repeat measurements in the laboratory after storage in the earth's field showed a significant component of viscous magnetization to be present. Suspecting the presence of an unstable secondary magnetization the samples were thermally cleaned, up to 625°C. In most cases a stable end point was reached after heating to 600°C. The thermally demagnetized sample means are plotted in Fig. 3C with respect to the present horizontal and in Fig. 3D with respect to the palaeohorizontal. The thermal demagnetization experiments indicate that a stable and probably hematite component of magnetization has successfully been isolated. Demagnetization has decreased the scatter considerably and produced a magnetic direction consistent within and between sites. The direction is well away from the present or dipole field direction. With

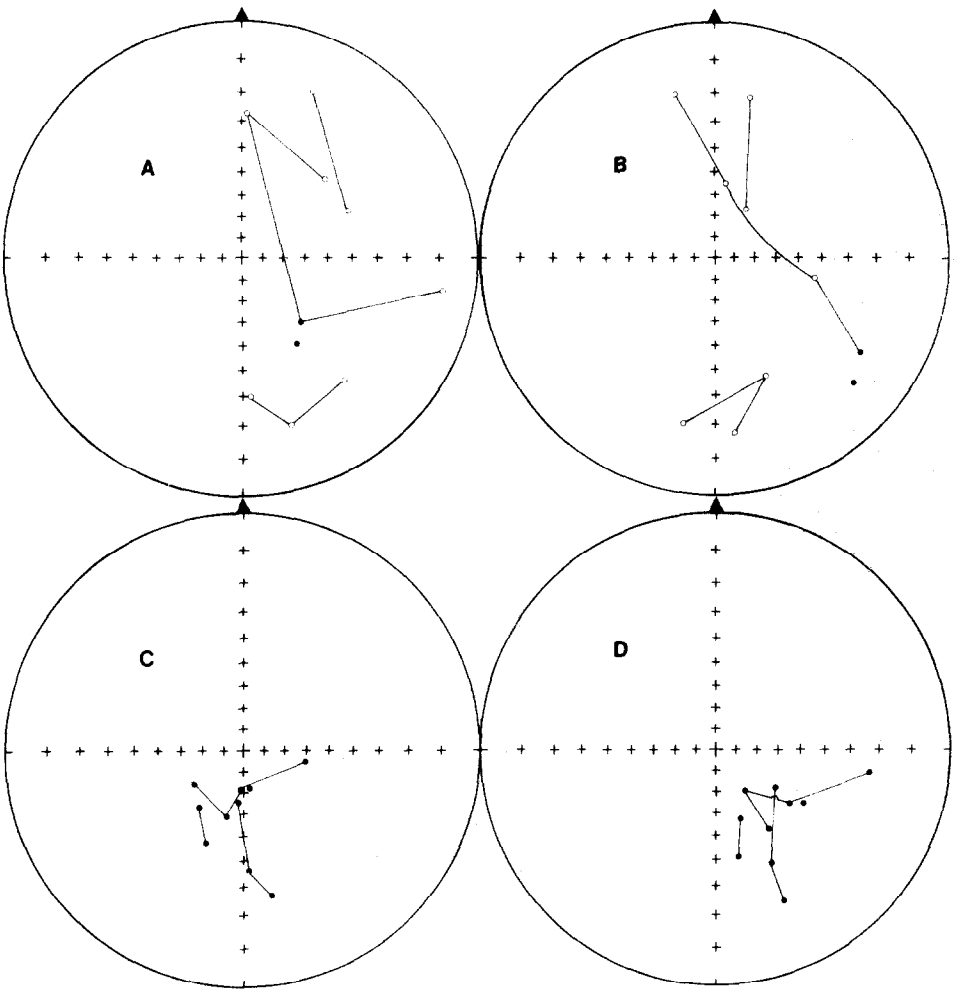



Fig. 3. Showing the directions of magnetization of the Maubisse red bed samples. A. N.M.R. directions with respect to present horizontal. B. N.R.M. direction with respect to bedding plane. C. Thermally cleaned directions with respect to horizontal. D. Thermally cleaned directions with respect to bedding plane. Lines connect samples from the same site.

respect to the palaeohorizontal the mean direction is found to be: Declination = 140.04, inclination = 45.02 with $\alpha_{95}$ = 13.06 and $N$ = 10 (samples). This corresponds to a pole position at 185.04° longitude and −48.06° latitude.

Application of the fold test according to McElhinny (1964) shows no significant improvement in the dispersion. This is largely due to insufficient variation in the bedding plane attitudes.

DISCUSSION

Despite the low sampling density and the obvious need to extend the collection over sites with significantly different bedding planes, so as to obtain a positive fold test, the results presented do suggest that a stable and ancient remanent magnetization for the Maubisse Formation has been isolated. It is of considerable interest to consider its implications.

In Fig. 4 are shown the directions of magnetization reported earlier (Chamalaun, 1977) for the Cribas Formation. The directions are with respect to the bedding plane and in this case a positive fold test was obtained. The presence of normal and reversed polarities and the consistency between volcanic and red beds attest to the reliability of the Cribas palaeomagnetic result. The Cribas Formation which occurs some 30 kilometres to the east of Maubisse, in the Cribas Anticline, is Late Permian in age and was mapped by Audley-Charles (1968) as wholly autochthonous. The pole position for the Cribas was compared with the Permo-Triassic poles for Australia (Chamalaun,

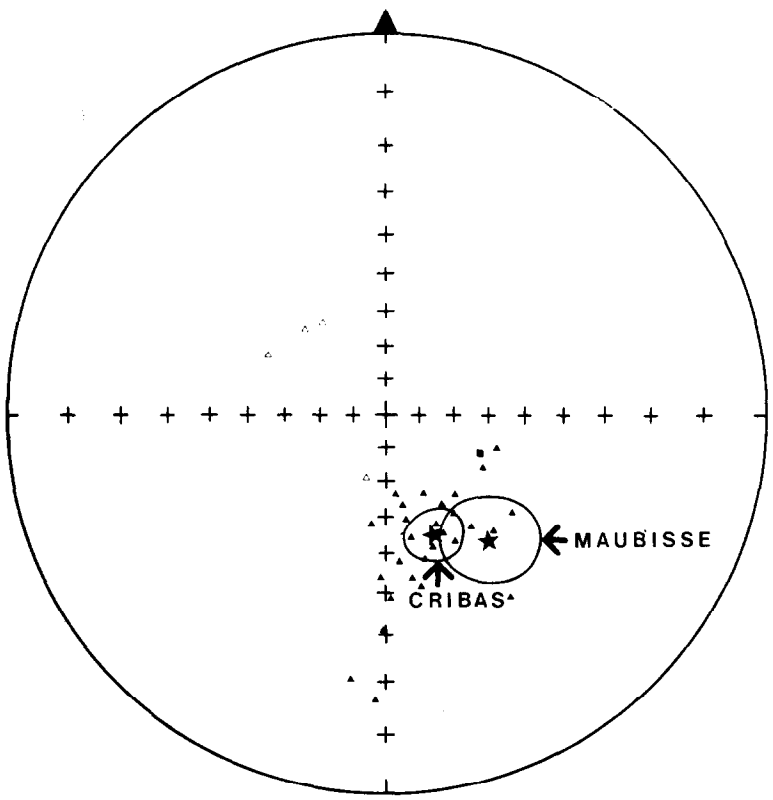



Fig. 4. Showing the directions of magnetization of the Cribas Formations with respect to the bedding plane (Chamalaun, 1977) after thermal demagnetization, and the error circles ($\alpha_{95}$) for the Cribas and Maubisse mean directions.

1977) and found to be in good agreement. This suggested that no significant relative displacement had occurred between Australia and autochthonous Timor, since the Permian. Thus the Cribas result is consistent with the hypothesis that autochthonous Timor formed part of the Australian continental margin since that time.

Comparing Fig. 3D and Fig. 4 shows that the Maubisse magnetization is not significantly different from that of the Cribas Formation as the error circles ($\alpha_{95}$) overlap and could well be regarded as identical. Since both Formations are of comparable age, it is difficult to avoid the conclusion that they were magnetized at the same time and in the same field. It then follows that the Maubisse Formation is not allochthonous, but part of autochthonous Timor. This conclusion is consistent with Grady's mapping of the 'Maubisse Thrust Plane' (Grady, 1975) and his more recent field mapping just west of Cribas, where it is found that the Cribas and Maubisse Formation are interfingered (Grady and Berry, 1976). The possibility that the Cribas and Maubisse Formations may have to be considered as equivalent cannot now be dismissed. However, confirmation needs to be obtained from further geological mapping and more extensive palaeomagnetic studies.

The apparent paradox between the Permian glaciations in northern Australia and the warm water fauna of Timor, has been an important element in the hypothesis of an allochthonous origin of the Maubisse and has led to the suggestion that the Maubisse was derived from some thousands of miles north of Timor. If the Maubisse were to be regarded as autochthonous and therefore part of the Australian continental margin, the paradox requires an alternative explanation. Such an explanation could emerge from one or a combination of the following considerations. The palaeolatitude indicated by the inclination of the Maubisse palaeomagnetic direction is 26°, which is not inconsistent with palaeolatitude distribution of the traditional warm water indicators such as red beds, coral and evaporates (Briden and Irving, 1963). Furthermore Runnegar and Campbell (1976), in discussing the Permo-Carboniferous faunas of Australia point out the presence of cold-water fauna in the Boreal and Gondwana zones, separated by a warm water Tethys fauna. The southern transition of the Tethys fauna to the Gondwana fauna in Australia is identified by them in the Bonaparte gulf region, some 600 kilometres S.E. of present-day Timor. It is quite conceivable therefore that the northern margin of Gondwanaland, i.e. Timor, was situated at the edge of the warm waters of the Tethys. A further consideration is that of time equivalence. The north Australian glacial deposits are probably Sakmarian in age (Crowell and Frakes, 1971) whereas the Maubisse and Cribas, are more likely to be Late Permian. These considerations strongly suggest that the paradox may be more apparent than real and certainly make the search for a 'far away' origin of the Maubisse unnecessary.

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